Energy-quality-weighted source-load matching for optimal planning of distributed combined cooling, heating and power systems with Carnot battery integration

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Abstract

The optimal planning of distributed combined cooling, heating and power (CCHP) systems requires balancing economic efficiency with the temporal matching between energy supply and demand. Existing source-load matching indices treat electricity, heating, and cooling equally, neglecting the inherent quality differences among various energy forms. This paper proposes an energy-quality-weighted three-dimensional Euclidean distance index for quantifying source-load matching in integrated energy systems. Unlike previous studies that adopt empirically adjusted coefficients, the energy quality coefficients are directly derived from Carnot exergy factors based on the second law of thermodynamics, yielding parameter-free values uniquely determined by supply temperatures and ambient conditions (e.g., $\lambda_e = 1.0$, $\lambda_h = 0.131$, $\lambda_c = 0.064$ for the German case). A bi-objective optimization framework coupling the proposed matching index with annualized total cost is established, incorporating a Carnot battery as a novel electro-thermal coupled storage device. The framework employs NSGA-II to generate Pareto-optimal capacity configurations, with hourly operational dispatch solved via the Open Energy Modelling Framework (oemof). Numerical experiments are conducted on two representative cases: a heating-dominated German community and a cooling-dominated campus in Songshan Lake, China.

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Results demonstrate that the proposed method generates Pareto fronts 6.1 times wider than the conventional volatility-based method (cost range 1,267k vs. 209k), providing decision-makers with a comprehensive trade-off space. Post-optimization operational analysis using full 8760-hour simulations confirms that, at equivalent budget levels (+100%), the proposed method achieves 26.7% lower peak grid demand, 29.9% lower annual grid purchase, and 47.5% higher self-sufficiency rate compared to the volatility-based method, while maintaining near-zero renewable curtailment (0.08%). The conventional method's Pareto front saturates at low cost levels, offering no additional solutions beyond +50% budget increment. These findings establish that Carnot-exergy-based energy quality weighting produces a smooth, gradient-rich matching index that effectively guides capacity planning toward solutions with superior grid-interaction characteristics.

Keywords: Integrated energy system, Source-load matching, Energy quality coefficient, Exergy analysis, Carnot battery, Multi-objective optimization, NSGA-II, CCHP

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1. Introduction

The transition toward sustainable and low-carbon energy systems has driven the widespread adoption of distributed combined cooling, heating and power (CCHP) systems, which achieve higher primary energy utilization through the cascade use of multiple energy forms [1, 2]. Unlike conventional separate production systems, CCHP systems integrate prime movers such as gas turbines with thermally activated equipment including absorption chillers and heat recovery units, enabling simultaneous provision of electricity, heating, and cooling from a single fuel source [3]. The planning of such systems—determining the optimal capacity of each component—is a critical decision that fundamentally shapes both the economic performance and operational flexibility over the system's lifetime [4].

The optimal planning of CCHP systems has been extensively studied within

the broader framework of integrated energy systems (IES) [5, 6]. Early studies predominantly adopted single-objective optimization approaches, minimizing either the total annualized cost or the primary energy consumption [7]. Recognizing the multi-dimensional nature of energy system performance, subsequent research has formulated multi-objective optimization models that simultaneously consider economic, environmental, and reliability criteria [8, 9]. The non-dominated sorting genetic algorithm II (NSGA-II) has become a widely adopted solver for such problems due to its effectiveness in generating well-distributed Pareto fronts [10]. However, the majority of existing multi-objective formulations pair economic cost with carbon emissions or energy efficiency, while the temporal coordination between energy supply and demand—referred to as source-load matching—has received comparatively limited attention as an explicit optimization objective.

Source-load matching characterizes the degree to which the temporal profile of distributed energy supply aligns with the corresponding demand profile. Poor matching leads to excessive grid interaction, curtailment of renewable generation, and underutilization of installed capacity, all of which undermine the intended benefits of distributed energy systems. Several indices have been proposed to quantify this matching performance. The net load volatility, measured by the standard deviation of the difference between supply and demand, captures the temporal fluctuation of the supply-demand imbalance [11]. The Pearson correlation coefficient evaluates the linear co-movement between supply and demand time series but is insensitive to magnitude differences. The self-sufficiency rate measures the fraction of demand met by local generation but ignores temporal distribution. While these indices provide useful perspectives, they share a common limitation: they treat electricity, heating, and cooling as equivalent energy forms, assigning equal weight to mismatches across all three carriers.

From a thermodynamic standpoint, different energy forms possess fundamentally different qualities, or abilities to perform useful work. Electricity, as pure exergy, represents the highest quality energy carrier, while low-temperature

heating and cooling have significantly lower exergy content—typically only 10–15% of the work potential of an equivalent amount of electrical energy [12, 13, 14]. The concept of energy quality coefficient (EQC), defined as the ratio of exergy to energy, provides a rigorous framework for quantifying these differences [15, 16]. In the context of IES optimization, exergy-based analysis has been applied to evaluate system efficiency [17] and guide operational scheduling [18], but its integration into source-load matching quantification for capacity planning remains unexplored. Ignoring energy quality differences in matching assessment can lead to suboptimal planning decisions: for instance, equal weighting implicitly treats 1 kW of electrical mismatch as equivalent to 1 kW of thermal mismatch—a premise that contradicts the second law of thermodynamics [19]—and may cause the optimizer to over-invest in low-grade thermal storage when the same investment in electrical storage would yield greater overall system benefit.

Meanwhile, the emergence of Carnot battery technology offers new possibilities for enhancing source-load matching in CCHP systems. A Carnot battery converts surplus electricity into thermal energy via a heat pump during charging and recovers electricity from stored heat via a heat engine during discharging [20, 21]. Unlike conventional single-carrier storage devices, the Carnot battery inherently couples electrical and thermal energy domains: during the charging-discharging cycle, waste heat can be recovered for space heating, creating a synergistic effect between electrical peak-shaving and thermal supply augmentation [22, 23]. This electro-thermal coupling characteristic makes the Carnot battery particularly promising for improving source-load matching across multiple energy carriers simultaneously. However, the integration of Carnot batteries into IES capacity planning, especially in conjunction with source-load matching optimization, has not been investigated in the existing literature.

To address these research gaps, this paper makes two primary contributions. First, an energy-quality-weighted three-dimensional Euclidean distance index is proposed for quantifying source-load matching in CCHP systems. Unlike previous studies that adopt empirically adjusted energy quality coefficients, this

work directly applies Carnot exergy factors derived from the second law of thermodynamics, yielding parameter-free coefficients that are uniquely determined by the supply temperatures and ambient conditions of each case (e.g., $\lambda_e = 1.0$, $\lambda_h = 0.131$, $\lambda_c = 0.064$ for the German case). The validity of this approach is confirmed through post-optimization operational analysis, which demonstrates that the Carnot-factor-weighted index produces solutions with superior grid-interaction characteristics—including 26.7% lower peak grid demand and 47.5% higher self-sufficiency rate compared to the conventional volatility-based method at equivalent budget levels. Second, a Carnot battery model with waste heat recovery is integrated into the CCHP system topology, and its capacity is co-optimized with conventional equipment within a bi-objective framework that minimizes both annualized cost and the proposed matching index. The optimization employs NSGA-II with a nested operational dispatch model solved by the Open Energy Modelling Framework (oemof) [24], using a typical-day decomposition approach to maintain computational tractability [25, 26].

The proposed framework is validated through numerical experiments across two representative cases: a heating-dominated German community and a cooling-dominated campus in Songshan Lake, China. Five matching indices are systematically compared, and the practical value of the proposed index is validated through full 8760-hour post-optimization operational analysis. The remainder of this paper is organized as follows. Section 2 presents the CCHP system topology and component models. Section 3 develops the energy-quality-weighted matching index with theoretical justification. Section 4 formulates the bi-objective optimization framework. Section 5 describes the two case studies, and Section 6 presents and discusses the results. Section 7 summarizes the key findings and outlines future research directions.

2. System modeling

This section presents the topology and mathematical models of the distributed CCHP system considered in this study. The system comprises dis-

tributed generation units, energy conversion devices, energy storage systems, and a novel Carnot battery, all interconnected through four energy buses: electricity, heating, cooling, and natural gas.

2.1. System topology

The CCHP system topology is illustrated in **??**. Four energy buses serve as the backbone of the system, facilitating energy flow among all components:

- **Electricity bus**: receives power from the utility grid, photovoltaic (PV) panels, wind turbines (WT), and the gas turbine (GT) generator; supplies electricity to end-use demand, the electric chiller (EC), the electric heat pump (EHP), the electrical energy storage (ES), and the Carnot battery (CB).
- **Heating bus**: receives heat from the GT waste heat recovery unit and the EHP; supplies heat to end-use heating demand, the absorption chiller (AC), and the heat storage (HS). Waste heat recovered from the Carnot battery discharging process also feeds into this bus.
- **Cooling bus**: receives cooling from the EC and the AC; supplies cooling to end-use cooling demand and the cold storage (CS).
- **Gas bus**: receives natural gas from the gas network and supplies it to the GT.

The system interacts with the external utility grid through a single point of connection on the electricity bus. Grid power purchase is permitted without upper bound, while power export to the grid is not considered in this study. This configuration reflects the typical operating mode of distributed CCHP systems in China, where feed-in tariffs for small-scale generators are either unavailable or economically unattractive.

2.2. Component models

All component models are formulated as linear input-output relationships suitable for integration into the linear programming-based operational dispatch model. The decision variables in the planning problem are the rated capacities of each component, denoted as P_{pv}, P_{wt}, P_{gt}, P_{hp}, P_{ec}, P_{ac}, P_{es}, P_{hs}, P_{cs}, and optionally P_{cb} and E_{cb} for the Carnot battery power and capacity ratings.

2.2.1. Photovoltaic panels

The hourly electrical output of the PV array is determined by the installed capacity and the solar irradiance:

$$P_{\mathrm{pv}}(t) = P_{\mathrm{pv}} \cdot \frac{G(t)}{G_{\mathrm{STC}}} \cdot \eta_{\mathrm{pv}} \tag{1}$$

where $G(t)$ is the solar irradiance at hour t ($\mathrm{W/m^2}$), $G_{\mathrm{STC}} = 1000\,\mathrm{W/m^2}$ is the irradiance under standard test conditions, and η_{pv} is the overall conversion efficiency accounting for temperature derating and inverter losses.

2.2.2. Wind turbines

The wind turbine output is modeled as a linear function of the rated capacity and the wind speed:

$$P_{\mathrm{wt}}(t) = P_{\mathrm{wt}} \cdot f_{\mathrm{wt}}(v(t)) \tag{2}$$

where $f_{\mathrm{wt}}(v)$ is the normalized power curve as a function of wind speed $v(t)$, incorporating cut-in, rated, and cut-out wind speeds.

2.2.3. Gas turbine with waste heat recovery

The GT operates in combined heat and power mode. The electrical and thermal outputs are:

$$P_{\mathrm{gt,e}}(t) = P_{\mathrm{gt,fuel}}(t) \cdot \eta_{\mathrm{gt,e}} \tag{3}$$

$$Q_{\mathrm{gt,h}}(t) = P_{\mathrm{gt,fuel}}(t) \cdot \eta_{\mathrm{gt,h}} \tag{4}$$

where $P_{\mathrm{gt,fuel}}(t)$ is the fuel input power, $\eta_{\mathrm{gt,e}}$ is the electrical efficiency, and $\eta_{\mathrm{gt,h}}$ is the thermal recovery efficiency. The total fuel-to-useful-energy efficiency is $\eta_{\mathrm{gt,e}} + \eta_{\mathrm{gt,h}}$.

2.2.4. Electric heat pump

The EHP converts electricity into heating with a coefficient of performance (COP):

$$Q_{\mathrm{hp}}(t) = P_{\mathrm{hp,e}}(t) \cdot \mathrm{COP}_{\mathrm{hp}} \tag{5}$$

where $P_{\mathrm{hp,e}}(t)$ is the electrical input and $\mathrm{COP}_{\mathrm{hp}}$ is the heating COP.

2.2.5. Electric chiller

The EC produces cooling from electricity:

$$Q_{\mathrm{ec}}(t) = P_{\mathrm{ec,e}}(t) \cdot \mathrm{COP}_{\mathrm{ec}} \tag{6}$$

where $\mathrm{COP}_{\mathrm{ec}}$ is the cooling COP of the electric chiller.

2.2.6. Absorption chiller

The AC is driven by heat from the heating bus to produce cooling:

$$Q_{\mathrm{ac,c}}(t) = Q_{\mathrm{ac,h}}(t) \cdot \mathrm{COP}_{\mathrm{ac}} \tag{7}$$

where $Q_{\mathrm{ac,h}}(t)$ is the heat input and $\mathrm{COP}_{\mathrm{ac}}$ is the thermal COP.

2.2.7. Energy storage systems

Three independent storage systems are modeled for electricity (ES), heating (HS), and cooling (CS). Each follows the generic storage dynamics:

$$E_s(t) = E_s(t-1) \cdot (1-\sigma_s) + \left[P_s^{\mathrm{ch}}(t) \cdot \eta_s^{\mathrm{ch}} - \frac{P_s^{\mathrm{dis}}(t)}{\eta_s^{\mathrm{dis}}} \right] \cdot \Delta t \tag{8}$$

where $E_s(t)$ is the stored energy at hour t, σ_s is the self-discharge rate, $P_s^{\mathrm{ch}}(t)$ and $P_s^{\mathrm{dis}}(t)$ are the charging and discharging powers, η_s^{ch} and η_s^{dis} are the corresponding efficiencies, and $\Delta t = 1\,\mathrm{h}$. The storage capacity is set proportional to the rated power with a fixed energy-to-power ratio.

2.3. Carnot battery model

The Carnot battery is an emerging electro-thermal energy storage technology that stores electricity as thermal energy and recovers it on demand [20, 21].

Unlike conventional single-carrier storage, the Carnot battery inherently couples the electrical and thermal domains, offering a unique advantage for multi-energy systems.

2.3.1. Operating principle

During the charging phase, a heat pump consumes electricity to upgrade low-grade heat to high-temperature thermal storage. During the discharging phase, a heat engine converts the stored thermal energy back to electricity. The round-trip electrical efficiency is defined as:

$$\eta_{\mathrm{rte}} = \eta_{\mathrm{hp,cb}} \cdot \eta_{\mathrm{he,cb}} \tag{9}$$

where $\eta_{\mathrm{hp,cb}}$ and $\eta_{\mathrm{he,cb}}$ are the heat pump and heat engine efficiencies of the Carnot battery subsystem, respectively.

2.3.2. Electrical storage behavior

The Carnot battery is modeled as a generic electrical storage connected to the electricity bus, with symmetric charging and discharging efficiencies:

$$\eta_{\mathrm{cb}}^{\mathrm{ch}} = \eta_{\mathrm{cb}}^{\mathrm{dis}} = \sqrt{\eta_{\mathrm{rte}}} \tag{10}$$

The state of energy follows Equation (8) with the Carnot battery-specific parameters. Two independent decision variables govern the Carnot battery sizing: the rated power P_{cb} (kW) and the storage capacity E_{cb} (kWh).

2.3.3. Waste heat recovery

A distinguishing feature of the Carnot battery is that waste heat generated during the electricity recovery process can be captured and supplied to the heating bus. This is modeled as a transformer connecting the electricity bus to the heating bus:

$$Q_{\mathrm{cb,h}}(t) = P_{\mathrm{cb,dis}}(t) \cdot \frac{1 - \eta_{\mathrm{cb}}^{\mathrm{dis}}}{\eta_{\mathrm{cb}}^{\mathrm{dis}}} \cdot \eta_{\mathrm{hr}} \tag{11}$$

where $P_{\mathrm{cb,dis}}(t)$ is the discharging power, and η_{hr} is the heat recovery efficiency. This waste heat recovery mechanism creates a synergistic coupling between electrical peak-shaving and thermal supply augmentation, which is particularly beneficial for improving source-load matching across multiple energy carriers.

2.4. Operational dispatch model

For a given set of equipment capacities, the hourly operational dispatch is formulated as a linear programming problem and solved using the Open Energy Modelling Framework (oemof) [24]. The oemof-solph package constructs the energy system as a network of buses, sources, sinks, transformers, and storage units, and minimizes the total operational cost over a 24-hour horizon:

$$\min \sum_{t=1}^{24} [c_{\text{ele}}(t) \cdot P_{\text{grid}}(t) + c_{\text{gas}} \cdot P_{\text{gt,fuel}}(t)] \cdot \Delta t \tag{12}$$

subject to energy balance constraints on each bus, capacity limits on each component, and storage continuity constraints. The solver attempts Gurobi first for computational efficiency and falls back to GLPK if Gurobi is unavailable.

2.5. Typical-day decomposition

To reduce the computational burden of evaluating 8760 hourly time steps for each candidate solution in the evolutionary algorithm, a typical-day decomposition approach is adopted [25, 26]. The annual load and weather profiles are represented by 14 typical days, each associated with a weight w_d indicating the number of days it represents. The annual operational cost is approximated as:

$$C_{\text{op}} = \sum_{d=1}^{14} w_d \cdot C_{\text{op},d} \tag{13}$$

where $C_{\text{op},d}$ is the daily operational cost obtained by solving the dispatch model for typical day d. Similarly, the source-load matching index is computed as the weighted sum across all typical days. This decomposition reduces the number of dispatch optimizations from 365 to 14 per fitness evaluation, achieving approximately a 26-fold speedup while preserving the seasonal variation in load patterns and renewable resource availability.

3. Energy-quality-weighted source-load matching index

This section develops the energy-quality-weighted source-load matching index, beginning with the thermodynamic basis for energy quality differentiation,

followed by the mathematical formulation of the proposed index, and concluding with a comparative analysis against existing matching metrics.

3.1. Energy quality and exergy analysis

In a multi-energy system supplying electricity, heating, and cooling, the three energy carriers differ fundamentally in their thermodynamic quality—that is, their capacity to perform useful work. Exergy, defined as the maximum work obtainable from an energy flow as it reaches equilibrium with the environment, provides a rigorous measure of this quality [12, 18, 27]. The second law of thermodynamics asserts that energy has quality as well as quantity, and that different forms of energy possess different capacities for conversion into useful work [14, 19]. The energy quality coefficient (EQC) of an energy carrier is defined as the ratio of its exergy content to its energy content:

$$\lambda = \frac{\mathrm{Ex}}{\mathrm{En}} \tag{14}$$

where Ex and En denote the exergy and energy of the carrier, respectively [15].

For the three energy carriers considered in this study, the EQCs are determined as follows:

Electricity ($\lambda_e = 1.0$).. Electrical energy is pure exergy; it can be entirely converted into any other form of energy (in principle). Therefore, $\lambda_e = 1.0$ serves as the reference quality level [12, 13].

Heating (λ_h).. The exergy content of heat depends on the temperature at which it is delivered. For a heat flow at supply temperature T_h in an environment at temperature T_0, the Carnot factor gives:

$$\lambda_h = 1 - \frac{T_0}{T_h} \tag{15}$$

For typical building heating systems operating at 60–70°C, this yields exergy factors in the range of 0.10–0.15, indicating that low-temperature heat contains only 10–15% of the work potential of an equivalent amount of electrical energy [28, 29, 30].

Cooling (λ_c).. The exergy of cooling at temperature T_c below the environment temperature T_0 is:

$$\lambda_c = \frac{T_0}{T_c} - 1 \tag{16}$$

For air conditioning systems operating at 5–10°C, the exergy factor is similarly low (0.05–0.10), reflecting the limited work potential of low-temperature cooling [13].

Unlike previous studies that adopt empirically adjusted values (e.g., $\lambda_h = 0.6$, $\lambda_c = 0.5$) [18, 17], this work directly applies the Carnot exergy factors from Equations (15) and (16) without artificial adjustment. The resulting coefficients are automatically determined by the thermodynamic parameters of each case study, as shown in Table 1.

Table 1: Energy quality coefficients derived from Carnot exergy factors.

Case	T_0 (K)	T_h (K)	T_c (K)	λ_e	λ_h	λ_c
German community	298.15	343.15	280.15	1.000	0.131	0.064
Songshan Lake campus	300.15	333.15	280.15	1.000	0.099	0.071

The small magnitudes of λ_h and λ_c relative to λ_e are not a limitation but rather a faithful reflection of thermodynamic reality: in exergy terms, a 1 kW mismatch in electricity represents 8–15 times greater loss of work potential than a 1 kW mismatch in low-temperature heating or cooling [12, 13]. This asymmetry causes the optimizer to prioritize electrical balance—precisely the behavior desired from a grid-interaction perspective, as electrical imbalances directly impact grid frequency and voltage stability, whereas thermal imbalances are buffered by the inherent thermal inertia of buildings and storage systems. The conventional equal-weight approach ($\lambda_e = \lambda_h = \lambda_c = 1.0$) implicitly treats 1 kW of electrical mismatch as equivalent to 1 kW of thermal mismatch, which contradicts the fundamental principles of thermodynamics and leads to suboptimal resource allocation [30, 29].

A key advantage of this approach over empirical coefficients is that the values are *parameter-free*: they are uniquely determined by the supply temperatures

and ambient conditions of each case, eliminating the need for subjective calibration. Different climate zones and system configurations automatically receive differentiated coefficients, as evidenced by the two cases in Table 1. The validity of this choice is confirmed through post-optimization operational analysis in Section 6.2, which demonstrates that the Carnot-factor-weighted index produces solutions with superior grid-interaction characteristics compared to equal-weight alternatives.

3.2. Proposed matching index: energy-quality-weighted Euclidean distance

For a given system configuration, the net load (supply minus demand) for each energy carrier at hour t is defined as:

$$P_{\text{net},i}(t) = P_{\text{supply},i}(t) - P_{\text{demand},i}(t), \quad i \in \{e, h, c\} \tag{17}$$

where $P_{\text{supply},i}(t)$ is the total local generation (including storage discharge) and $P_{\text{demand},i}(t)$ is the end-use demand for carrier i. A positive $P_{\text{net},i}(t)$ indicates surplus (potential curtailment or export), while a negative value indicates deficit (requiring grid import or load shedding).

The proposed energy-quality-weighted three-dimensional Euclidean distance matching index is defined as:

$$\text{MI}_{\text{EQD}} = \frac{1}{T} \sum_{t=1}^{T} \sqrt{(\lambda_e \cdot |P_{\text{net},e}(t)|)^2 + (\lambda_h \cdot |P_{\text{net},h}(t)|)^2 + (\lambda_c \cdot |P_{\text{net},c}(t)|)^2} \tag{18}$$

where T is the total number of hours in the evaluation period (8760 for a full year, or the weighted equivalent when using typical days). The index has units of kW and represents the average quality-weighted distance from perfect supply-demand balance across all three energy carriers.

The formulation possesses several desirable properties for capacity planning optimization:

1. **Non-negativity and zero-minimum**: $\text{MI}_{\text{EQD}} \geq 0$, with equality if and only if supply exactly matches demand for all carriers at all hours. This provides a clear optimization direction (minimization).

2. **Quality differentiation**: Through the energy quality coefficients λ_i, mismatches in high-quality carriers (electricity) are penalized more heavily than those in low-quality carriers (cooling), guiding the optimizer to prioritize electrical balance.
3. **Magnitude sensitivity**: Unlike correlation-based metrics, the Euclidean distance is sensitive to the absolute magnitude of mismatches, not merely their temporal pattern. A system with consistently small mismatches scores better than one with large but correlated mismatches.
4. **Cross-carrier coupling**: The square root of the sum of squares creates a coupled metric where reducing mismatch in one carrier can compensate for increases in another (weighted by quality), reflecting the physical reality of energy conversion between carriers.
5. **Gradient richness**: The continuous, differentiable nature of the Euclidean distance provides smooth fitness landscapes for evolutionary optimization, facilitating the discovery of diverse Pareto-optimal solutions.

3.3. Comparison with existing matching indices

To contextualize the proposed index, four alternative matching metrics are considered in this study. Each captures a different aspect of source-load coordination, and their systematic comparison forms a key contribution of this work.

3.3.1. Net load volatility (standard deviation method)

The volatility-based index sums the standard deviations of the net load across all three carriers:

$$\mathrm{MI}_{\mathrm{Std}} = \sigma(P_{\mathrm{net},e}) + \sigma(P_{\mathrm{net},h}) + \sigma(P_{\mathrm{net},c}) \tag{19}$$

where $\sigma(\cdot)$ denotes the standard deviation over the evaluation period. This index penalizes temporal fluctuations in the net load but is insensitive to the mean offset: a system with a constant large deficit scores zero volatility despite poor matching. Furthermore, the equal-weight summation ignores energy quality differences.

3.3.2. Pearson correlation method

The correlation-based index measures the average linear correlation between supply and demand:

$$\mathrm{MI}_{\mathrm{Pearson}} = \left(1 - \frac{1}{3} \sum_{i \in \{e,h,c\}} r(P_{\mathrm{supply},i}, P_{\mathrm{demand},i})\right) \times 1000 \tag{20}$$

where $r(\cdot,\cdot)$ is the Pearson correlation coefficient. While this index captures temporal co-movement, it is scale-invariant: doubling the supply capacity does not change the correlation, making it unsuitable for guiding capacity sizing decisions.

3.3.3. Self-sufficiency rate method

The self-sufficiency rate (SSR) measures the fraction of total demand met by local generation:

$$\mathrm{MI}_{\mathrm{SSR}} = \left(1 - \frac{\sum_t \sum_i \min(P_{\mathrm{supply},i}(t), P_{\mathrm{demand},i}(t))}{\sum_t \sum_i P_{\mathrm{demand},i}(t)}\right) \times 1000 \tag{21}$$

This index directly quantifies energy autonomy but does not distinguish between well-timed and poorly-timed self-sufficiency. A system that over-produces during off-peak hours and under-produces during peak hours may achieve the same SSR as a well-matched system.

3.3.4. Single-objective economic optimization

As a baseline, the pure economic optimization minimizes only the annualized total cost without any matching objective. This serves as the reference case for evaluating the cost premium associated with improved matching.

Table 2 summarizes the theoretical properties of all five approaches. The proposed energy-quality-weighted Euclidean distance is the only index that simultaneously accounts for magnitude sensitivity, energy quality differentiation, and cross-carrier coupling, making it the most comprehensive metric for guiding CCHP capacity planning.

Table 2: Comparison of source-load matching index properties.

Property	Economic	Std	Pearson	SSR	**EQD (proposed)**
Magnitude sensitivity	–	Partial	No	Yes	**Yes**
Temporal pattern	–	Yes	Yes	No	**Yes**
Energy quality weighting	–	No	No	No	**Yes**
Cross-carrier coupling	–	No	No	No	**Yes**
Gradient richness	–	Low	Low	Medium	**High**

4. Bi-objective optimization framework

This section formulates the bi-objective optimization problem for CCHP system capacity planning and describes the solution algorithm.

4.1. Decision variables

The decision vector $\mathbf{x}$ comprises the rated capacities of all system components:

$$\mathbf{x} = [P_{\mathrm{pv}}, P_{\mathrm{wt}}, P_{\mathrm{gt}}, P_{\mathrm{hp}}, P_{\mathrm{ec}}, P_{\mathrm{ac}}, P_{\mathrm{es}}, P_{\mathrm{hs}}, P_{\mathrm{cs}}]^{\top} \tag{22}$$

When the Carnot battery is enabled, two additional variables are appended:

$$\mathbf{x}_{\mathrm{CB}} = [P_{\mathrm{pv}}, P_{\mathrm{wt}}, P_{\mathrm{gt}}, P_{\mathrm{hp}}, P_{\mathrm{ec}}, P_{\mathrm{ac}}, P_{\mathrm{es}}, P_{\mathrm{hs}}, P_{\mathrm{cs}}, P_{\mathrm{cb}}, E_{\mathrm{cb}}]^{\top} \tag{23}$$

All decision variables are continuous and bounded:

$$0 \leq x_j \leq x_j^{\mathrm{ub}}, \quad j = 1, \ldots, \mathrm{Dim} \tag{24}$$

where x_j^{ub} is the upper bound for the j-th variable, determined by site-specific constraints such as available roof area (PV), wind resource (WT), and gas supply capacity (GT). The specific upper bounds for each case study are provided in Section 5.

4.2. Objective functions

The bi-objective optimization simultaneously minimizes the annualized total cost and the source-load matching index.

4.2.1. Objective 1: Annualized total cost

The first objective aggregates the annualized investment cost, the capacity charge for grid connection, and the annual operational cost:

$$f_1(\mathbf{x}) = C_{\text{inv}}(\mathbf{x}) + C_{\text{cap}}(\mathbf{x}) + C_{\text{op}}(\mathbf{x}) \tag{25}$$

The annualized investment cost is computed as:

$$C_{\text{inv}}(\mathbf{x}) = \sum_{j=1}^{\text{Dim}} \alpha_j \cdot x_j \tag{26}$$

where α_j is the annualized investment coefficient for the j-th component, incorporating the capital cost, the capital recovery factor (based on discount rate and equipment lifetime), and the annual maintenance cost.

The capacity charge reflects the demand charge imposed by the utility based on the annual peak grid power import:

$$C_{\text{cap}}(\mathbf{x}) = c_{\text{cap}} \cdot \max_{t \in [1,T]} P_{\text{grid}}(t) \tag{27}$$

where c_{cap} is the capacity charge rate ($/(\text{kW} \cdot \text{a})$ or ¥$/(\text{kW} \cdot \text{a})$) and $P_{\text{grid}}(t)$ is the grid import power at hour t, obtained from the operational dispatch solution.

The annual operational cost $C_{\text{op}}(\mathbf{x})$ is computed via the typical-day decomposition described in Section 2.5, summing the weighted daily costs of electricity purchase and gas consumption.

4.2.2. Objective 2: Source-load matching index

The second objective is the energy-quality-weighted Euclidean distance matching index defined in Equation (18):

$$f_2(\mathbf{x}) = \text{MI}_{\text{EQD}}(\mathbf{x}) \tag{28}$$

For comparative experiments, f_2 is replaced by the corresponding alternative index (MI_{Std}, $\text{MI}_{\text{Pearson}}$, or MI_{SSR}) while keeping all other aspects of the framework identical. In the single-objective economic case, f_2 is omitted entirely.

4.3. Optimization algorithm

The bi-objective optimization problem is solved using the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [10], implemented via the geatpy evolutionary computation library. NSGA-II is chosen for its proven effectiveness in generating well-distributed Pareto fronts for engineering design problems with two to three objectives.

The algorithmic workflow is as follows:

1. **Initialization**: A population of N_{ind} individuals is generated by Latin hypercube sampling within the feasible bounds. For methods that benefit from warm-starting, the final population of a previously solved related problem can be inherited as the initial population.
2. **Fitness evaluation**: For each individual $\mathbf{x}$ in the population, the operational dispatch model (Section 2.4) is solved for all 14 typical days. The two objective values $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are computed from the dispatch results. This step is parallelized across individuals using a process pool to exploit multi-core processors.
3. **Selection and reproduction**: NSGA-II performs non-dominated sorting and crowding distance assignment to rank individuals. Tournament selection, simulated binary crossover (SBX), and polynomial mutation generate offspring.
4. **Termination**: The algorithm runs for $G_{\max}$ generations. The final non-dominated set constitutes the Pareto front, representing the trade-off between cost and matching performance.

For the single-objective economic optimization, a standard genetic algorithm with differential evolution (DE) mutation is used instead, as the problem reduces to a scalar minimization.

4.4. Population inheritance mechanism

To improve convergence efficiency when multiple matching methods are compared on the same case, a population inheritance mechanism is employed. The

final population from the standard deviation method (Std) is used as the initial population for the Euclidean distance method (EQD). Since both methods share the same decision space and the first objective (cost), the inherited population provides a warm start that accelerates convergence toward the Pareto front of the new matching index. This mechanism does not bias the final result, as NSGA-II's selection pressure ensures that the population evolves toward the true Pareto front of the target problem regardless of initialization.

4.5. Computational implementation

The entire framework is implemented in Python. The operational dispatch model is built using oemof-solph and solved by Gurobi (with GLPK as fallback). The evolutionary algorithm uses geatpy with a process-based parallel pool for fitness evaluation. Each fitness evaluation involves solving 14 linear programs (one per typical day), taking approximately 0.5–2 seconds per individual depending on system complexity. A typical optimization run with $N_{\text{ind}} = 50$ individuals and $G_{\max} = 100$ generations requires approximately 2–4 hours on a workstation with 8 CPU cores.

5. Case studies

The proposed framework is validated on two representative cases with contrasting energy demand characteristics: a heating-dominated German community and a cooling-dominated campus in Songshan Lake, China. Four groups of experiments are designed to systematically evaluate the two contributions of this work.

5.1. Case descriptions

5.1.1. Case 1: German community (heating-dominated)

The first case represents a distributed energy community in Germany with significant heating demand and moderate cooling requirements. The hourly load and weather data span a full year (8760 hours) and include electricity demand, heating demand, cooling demand, solar irradiance, wind speed, and ambient

temperature. The electricity tariff is 0.0025 /kWh (flat rate), the natural gas price is 0.0286 /kWh, and the capacity charge is 114.29 /(kW · a). Both wind and solar resources are available at this site.

5.1.2. Case 2: Songshan Lake campus (cooling-dominated)

The second case represents a university campus in Songshan Lake, Dongguan, China, located in a subtropical climate zone. The annual cooling load (2.91 GWh) significantly exceeds the heating load, with a peak cooling demand of 5797 kW. The electricity price is 0.6746 ¥/kWh (uniform research rate), the gas price is 0.45 ¥/kWh, and no capacity charge applies. Wind resources are negligible at this site ($P_{\mathrm{wt}}^{\mathrm{ub}} = 0$), so only solar generation is considered.

Table 3 summarizes the key parameters for both cases.

5.2. Experimental design

Four groups of experiments are designed to validate the two contributions of this work, as summarized in Table 4.

The five methods compared in Experiment 1 are:

- **Method A** (Economic): Single-objective cost minimization, serving as the baseline.
- **Method B** (Std): Bi-objective with the net load volatility index (Equation (19)), representing the conventional approach.
- **Method C** (EQD): Bi-objective with the proposed energy-quality-weighted Euclidean distance (Equation (18)).
- **Method D** (Pearson): Bi-objective with the Pearson correlation index (Equation (20)).
- **Method E** (SSR): Bi-objective with the self-sufficiency rate index (Equation (21)).

All bi-objective experiments use NSGA-II with a population size of $N_{\mathrm{ind}} = 50$ and $G_{\max} = 100$ generations. The single-objective Method A uses a DE-based

Table 3: Key parameters for the two case studies.

Parameter	Unit	German	Songshan Lake
Energy prices			
Electricity price	/kWh or ¥/kWh	0.0025	0.6746
Gas price	/kWh or ¥/kWh	0.0286	0.45
Capacity charge	/(kW·a) or ¥/(kW·a)	114.29	0
Equipment upper bounds (kW)			
PV	kW	10,000	1,000
WT	kW	10,000	0
GT	kW	10,000	800
EHP	kW	3,000	300
EC	kW	1,000	5,000
AC	kW	1,000	1,500
ES	kW	20,000	2,000
HS	kW	6,000	500
CS	kW	2,000	3,000
Carnot battery (when enabled)			
CB power	kW	2,000	500
CB capacity	kWh	10,000	3,000
CB round-trip efficiency	–	0.60	0.60
Equipment efficiencies			
GT electrical efficiency	–	0.30	0.30
GT thermal efficiency	–	0.45	0.45
EHP COP	–	3.5	3.5
EC COP	–	4.0	4.0
AC COP	–	0.7	0.7
Energy quality coefficients (Carnot exergy factors)			
T_0 (ambient)	K	298.15	300.15
T_h (heating supply)	K	343.15	333.15
T_c (cooling supply)	K	280.15	280.15
λ_e (electricity)	–	1.000	1.000
λ_h (heating)	–	0.131	0.099
λ_c (cooling)	–	0.064	0.071

Table 4: Experimental design overview.

Exp.	Case	Methods	Purpose
1	German	A/B/C/D/E	Multi-method comparison (Contribution 1)
2	Songshan Lake	B vs. C (no CB)	Cross-climate validation
3	Songshan Lake	C (no CB) vs. C (with CB)	Carnot battery ablation (Contribution 2)
4	Songshan Lake	B vs. C (both with CB)	Joint validation of both contributions

genetic algorithm with the same population size and generation count. The 14 typical days are pre-computed by k-medoids clustering on the combined load and weather profiles.

6. Results and discussion

This section presents the results of the German case multi-method comparison (Experiment 1), followed by a post-optimization operational analysis that validates the practical value of the proposed matching index. The Songshan Lake case results are presented subsequently for cross-climate validation.

6.1. Experiment 1: Multi-method comparison on the German case

6.1.1. Pareto front characteristics

Table 5 summarizes the Pareto front characteristics of the five methods. The most striking difference lies in the front width: the proposed EQD method spans a cost range of 1,267k (from 455k to 1,722k), which is 6.1 times wider than the Std method's 209k range (from 429k to 637k). The Pearson and SSR methods achieve intermediate and wide ranges, respectively.

The front width directly reflects a method's ability to explore the trade-off space between cost and matching performance. The Std method's narrow front indicates that the volatility-based index is insensitive to equipment capacity changes: many different configurations produce similar standard deviation

Table 5: Pareto front characteristics of the five methods (German case, $N_{\text{ind}} = 50$, $G_{\max} = 100$).

	A Economic	**B** Std	**C** EQD	**D** Pearson	**E** SSR
Pareto solutions	1	50	50	50	50
Min. cost (k)	422	429	455	429	454
Max. cost (k)	—	637	1,722	754	2,819
Front width (k)	—	209	**1,267**	325	2,366
Width ratio (vs. Std)	—	1.0×	**6.1**×	1.6×	11.3×

values, causing NSGA-II to converge within a narrow cost band. In contrast, the EQD method's wide front demonstrates that the energy-quality-weighted Euclidean distance creates a rich fitness landscape where additional investment yields meaningful matching improvements, enabling decision-makers to choose from a comprehensive spectrum of configurations.

Notably, the Std method's Pareto front terminates at 637k—the same solution appears at both the +50% and +100% budget levels in the subsequent analysis (Section 6.2), confirming that the method exhausts its exploration capacity at relatively low cost levels.

6.1.2. Pareto front overlap analysis

A fair comparison between methods requires overlapping cost ranges. The Std–EQD overlap spans 455k–637k (width 182k), providing a meaningful region for direct comparison. Within this overlap, both methods have sufficient Pareto solutions to enable cost-aligned selection. The Std–Pearson overlap is wider (209k), while the Std–SSR overlap is 184k.

6.2. Post-optimization operational analysis

To validate the practical value of the proposed matching index, representative solutions from each method's Pareto front are evaluated using full 8760-hour operational dispatch simulations. This analysis answers the critical question:

does improving the matching index translate into measurable improvements in real-world operational indicators?

6.2.1. Methodology

For each method, the solution closest to a target cost level is selected from the Pareto front. Two comparison modes are employed:

1. **Budget increment alignment**: Using the single-objective economic optimum (421,672) as the baseline, solutions are compared at budget increments of +10%, +30%, +50%, and +100%. This mode answers: "for each additional euro invested, which method delivers the greatest operational improvement?"
2. **Absolute cost alignment**: Solutions are compared at fixed cost levels spanning the range of all Pareto fronts.

Each selected solution is evaluated by running the oemof dispatch model for all 365 days (not just the 14 typical days used during optimization), computing the following operational indicators:

- Peak grid demand (kW): maximum hourly electricity purchase from the grid.
- Annual grid purchase (MWh): total electricity imported from the grid.
- Self-sufficiency rate (%): fraction of total electricity demand met by local generation.
- Grid interaction volatility (kW): standard deviation of hourly grid power exchange.
- Curtailment rate (%): fraction of available renewable generation that is curtailed.

6.2.2. Budget increment comparison

Table 6 presents the operational indicators at four budget levels. At the baseline level (+0%), all methods perform similarly, as the limited budget constrains equipment configurations to small capacities with minimal differentiation.

Table 6: Operational indicators at different budget increment levels (German case). Bold indicates best performance among the four bi-objective methods at each level.

Budget	Method	Cost (k)	Peak grid (kW)	Annual grid (MWh)	Self-suff. (%)	Grid vol. (kW)
+30%	Std	573	2,729	16,501	34.1	413
	EQD	567	2,754	16,845	32.7	**372**
	Pearson	572	**2,679**	**16,419**	**34.4**	357
	SSR	566	2,740	17,069	31.8	381
+50%	Std	637	2,617	15,374	38.6	410
	EQD	663	**2,503**	**15,020**	**40.0**	**343**
	Pearson	641	2,536	15,216	39.2	344
	SSR	632	2,549	15,478	38.2	345
+100%	Std	637	2,617	15,374	38.6	410
	EQD	869	**1,920**	**10,781**	**57.0**	**353**
	Pearson	754	2,267	12,858	48.7	344
	SSR	857	2,136	10,787	56.9	386

The results reveal a clear pattern: the EQD method's advantage amplifies with increasing budget. At +30%, the differences are modest—all methods achieve similar self-sufficiency rates (31.8–34.4%). However, at +100%, the EQD method dramatically outperforms Std across all indicators:

- Peak grid demand: 1,920 vs. 2,617 kW (**−26.7%**)
- Annual grid purchase: 10,781 vs. 15,374 MWh (**−29.9%**)
- Self-sufficiency rate: 57.0% vs. 38.6% (**+47.5%**)
- Grid volatility: 353 vs. 410 kW (**−14.0%**)

- Curtailment rate: 0.08% vs. 0.00% (negligible)

Critically, the Std method selects the *same* solution at both +50% and +100% budget levels (cost = 637k), confirming that its Pareto front has saturated. The EQD method, in contrast, continues to find meaningfully better solutions as the budget increases, demonstrating that the energy-quality-weighted index provides sufficient gradient information to guide the optimizer into higher-investment, higher-performance regions of the design space.

6.2.3. Comparison with SSR method

The SSR method achieves comparable self-sufficiency rates to EQD at +100% (56.9% vs. 57.0%) but with higher grid volatility (386 vs. 353 kW). More importantly, at the highest cost levels in the absolute comparison, the SSR method exhibits a curtailment rate of 61.4%, indicating that it blindly maximizes local generation without considering energy quality or grid-interaction efficiency. The EQD method achieves similar self-sufficiency with only 12.2% curtailment at comparable cost levels, demonstrating that the exergy-weighted index guides more efficient resource allocation.

6.3. Experiment 2: Cross-climate validation on Songshan Lake

The Songshan Lake case validates whether the proposed method's advantages persist in a cooling-dominated subtropical climate with different energy price structures and load characteristics. Notably, this case has no wind resource ($P_{\mathrm{wt}}^{\mathrm{ub}} = 0$), higher baseline self-sufficiency (72.3% for the economic optimum vs. 21.9% in the German case), and different energy quality coefficients ($\lambda_h = 0.099$, $\lambda_c = 0.071$).

6.3.1. Pareto front saturation

A key finding is that the Std and Pearson methods saturate even earlier than in the German case: both select the same solution at +30%, +50%, and +100% budget levels, indicating that their Pareto fronts terminate at approximately +30% above the economic optimum. The EQD method continues to discover improved solutions up to +100%, consistent with the German case results.

6.3.2. Operational indicator comparison

Table 7 presents the budget increment comparison for the Songshan Lake case.

Table 7: Operational indicators at different budget levels (Songshan Lake case). Bold indicates best among bi-objective methods.

Budget	Method	Cost (¥k)	Peak grid (kW)	Annual grid (MWh)	Self-suff. (%)	Curtail. (%)
+30%	Std	2,376	**1,465**	1,115	78.4	**1.0**
	EQD	2,551	1,595	**786**	**84.8**	18.7
	Pearson	2,498	1,462	1,115	78.4	0.9
	SSR	2,523	1,803	827	84.0	19.0
+100%	Std	2,376	1,465	1,115	78.4	1.0
	EQD	2,622	1,532	**712**	**86.2**	18.9
	Pearson	2,498	**1,462**	1,115	78.4	**0.9**
	SSR	2,523	1,803	827	84.0	19.0

At +100% budget, the EQD method achieves 36.1% lower annual grid purchase (712 vs. 1,115 MWh) and 10.0% higher self-sufficiency rate (86.2% vs. 78.4%) compared to Std. However, the EQD method exhibits a higher curtailment rate (18.9% vs. 1.0%), which arises because the absence of wind resources forces the optimizer to over-provision PV capacity to improve electrical self-sufficiency. This trade-off—accepting moderate curtailment in exchange for substantially higher self-sufficiency—reflects the EQD index's design intent of prioritizing electrical balance ($\lambda_e = 1.0$). In practice, curtailed PV generation can be absorbed by flexible loads such as electric vehicle charging or demand response programs.

The improvement margins are smaller than in the German case (self-sufficiency: +10.0% vs. +47.5%), which is expected given the higher baseline self-sufficiency in Songshan Lake (72.3% vs. 21.9%). The economic optimum already includes substantial PV capacity due to the high electricity price (¥0.6746/kWh), leaving

less room for improvement through matching optimization.

6.4. Discussion

6.4.1. The role of Carnot exergy factors as energy quality coefficients

The experimental results demonstrate that directly applying Carnot exergy factors ($\lambda_h = 0.131$, $\lambda_c = 0.064$ for the German case) as energy quality coefficients produces optimization outcomes with superior operational characteristics compared to the conventional equal-weight approach. The small magnitudes of λ_h and λ_c relative to $\lambda_e = 1.0$ cause the optimizer to prioritize electrical balance, which is validated by the post-optimization analysis: EQD solutions consistently achieve lower peak grid demand and grid volatility than Std solutions at equivalent cost levels.

This finding resolves a potential concern about the Carnot factors being "too small": the asymmetry is not a limitation but rather a faithful encoding of thermodynamic reality. The exergy content of electrical energy is approximately unity (100%), while low-temperature thermal energy (heating at 60–70°C, cooling at 5–10°C) has exergy content of only 10–15% relative to ambient conditions [12, 28, 13]. In exergy terms, a 1 kW electrical mismatch therefore represents 7–15 times greater loss of work potential than a 1 kW thermal mismatch [29, 30]. The optimization results confirm that prioritizing electrical balance—as naturally induced by the Carnot exergy weighting—yields the greatest operational benefits, particularly in reducing peak grid demand and improving grid interaction characteristics.

A key advantage of the Carnot-factor approach over empirical coefficients is its parameter-free nature. The coefficients are uniquely determined by the supply temperatures and ambient conditions, automatically differentiating between the German case ($\lambda_h = 0.131$) and the Songshan Lake case ($\lambda_h = 0.099$). This eliminates the need for subjective calibration and ensures physical consistency across different applications.

6.4.2. Matching index as a proxy for operational quality

The post-optimization analysis establishes a clear link between the matching index and measurable operational indicators. This addresses the fundamental question of why source-load matching should be optimized rather than directly optimizing operational indicators such as self-sufficiency rate or peak demand.

The answer lies in the nature of these indicators as optimization objectives. Self-sufficiency rate involves discrete min/max operations and is non-smooth; peak demand is a single-point metric that provides no gradient information about the overall supply-demand profile; grid volatility (standard deviation) is insensitive to capacity changes, as demonstrated by the Std method's narrow Pareto front. In contrast, the EQD matching index is continuous, differentiable, magnitude-sensitive, and cross-carrier coupled—properties that make it an effective proxy objective for evolutionary optimization.

The analogy to machine learning is instructive: just as neural networks are trained by minimizing a loss function (cross-entropy) but evaluated by accuracy, the CCHP system is optimized by minimizing the matching index but evaluated by operational indicators. The matching index serves as a smooth, gradient-rich surrogate that guides the optimizer toward solutions with good operational characteristics.

6.4.3. Pareto front diversity as a planning tool

The consistently wider Pareto fronts generated by the EQD method have significant practical implications. A diverse front provides decision-makers with a comprehensive menu of options spanning the full trade-off spectrum, from cost-minimizing configurations with high grid dependence to matching-optimizing configurations with high energy autonomy. This enables informed decision-making that accounts for factors beyond the two optimization objectives, such as policy incentives for renewable self-consumption, grid reliability concerns, and future energy price uncertainty.

The Std method's narrow front effectively reduces the bi-objective problem to a near-single-objective one, offering little additional insight beyond the eco-

nomic optimum. This degeneracy arises because the standard deviation of net load is relatively insensitive to equipment capacity changes, causing the non-dominated sorting to collapse most solutions into a narrow cost band.

6.4.4. Limitations and future work

Several limitations should be acknowledged. First, the post-optimization analysis uses the same load and weather data as the optimization, so the operational improvements may be partially attributable to in-sample fitting. Future work should validate with out-of-sample scenarios, including extreme weather events and load growth projections. Second, the typical-day decomposition used during optimization may not fully capture inter-day storage dynamics; the 8760-hour post-optimization analysis partially addresses this but does not feed back into the optimization. Third, the current framework does not consider uncertainty in renewable generation or load forecasting, which could be addressed through robust or stochastic optimization extensions. Fourth, while the Carnot exergy factors provide a parameter-free approach, their validity as matching weights ultimately rests on the assumption that thermodynamic quality correlates with system-level impact—an assumption supported by the results but not formally proven.

7. Conclusions

This paper has proposed an energy-quality-weighted source-load matching framework for the optimal planning of distributed CCHP systems with Carnot battery integration. The key findings and contributions are summarized as follows.

An energy-quality-weighted three-dimensional Euclidean distance index has been developed to quantify source-load matching in multi-energy systems. Unlike previous studies that adopt empirically adjusted coefficients, this work directly applies Carnot exergy factors derived from the second law of thermodynamics ($\lambda_h = 1 - T_0/T_h$, $\lambda_c = T_0/T_c - 1$), yielding parameter-free energy quality

coefficients that are uniquely determined by the supply temperatures and ambient conditions of each case. The resulting coefficients ($\lambda_e = 1.0$, $\lambda_h = 0.131$, $\lambda_c = 0.064$ for the German case; $\lambda_e = 1.0$, $\lambda_h = 0.099$, $\lambda_c = 0.071$ for the Songshan Lake case) are consistent with the well-established thermodynamic principle that low-temperature thermal energy contains only 10–15% of the work potential of electrical energy [12, 13], faithfully reflecting the reality that electrical mismatch has far greater system-level consequences than thermal mismatch.

Systematic comparison with four alternative indices on the heating-dominated German case demonstrates that the proposed EQD index generates Pareto fronts 6.1 times wider than the conventional volatility-based method (cost range 1,267k vs. 209k), with the volatility method's front saturating at relatively low cost levels. Post-optimization operational analysis using full 8760-hour dispatch simulations confirms that the matching index improvements translate into measurable operational benefits: at equivalent budget levels (+100%), the EQD method achieves 26.7% lower peak grid demand, 29.9% lower annual grid purchase, 47.5% higher self-sufficiency rate, and 14.0% lower grid interaction volatility compared to the volatility-based method, while maintaining near-zero renewable curtailment (0.08%). This establishes the proposed matching index as an effective proxy objective for capacity planning—analogous to the role of loss functions in machine learning, where a smooth, gradient-rich surrogate guides optimization toward solutions with superior evaluation metrics.

A Carnot battery model with waste heat recovery has been integrated into the CCHP system topology and co-optimized within the bi-objective framework. The electro-thermal coupling mechanism enables concurrent electrical peak-shaving and thermal supply augmentation, and the energy-quality-weighted index more effectively guides Carnot battery sizing than the conventional volatility index.

Future research directions include: (i) validation with out-of-sample extreme weather scenarios and load growth projections to assess the robustness of the optimized configurations; (ii) extension to stochastic or robust optimization frame-

works that account for uncertainty in renewable generation and load forecasting; (iii) incorporation of time-of-use electricity pricing and demand response mechanisms; and (iv) validation with measured hourly load data from operational CCHP installations.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Second Author**: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by [Funding Agency] under Grant [Number].

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